UNIVERSITY OF THE PUNJAB

Roll No. ...CS8.937

B.S. 4 Years Program / Fifth Semester – Fall 2023

Time: 3 Hrs. Marks: 60

Paper: Advance Microeconomics Course Code: ECON-314

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short notes on the following: (6x5=30)

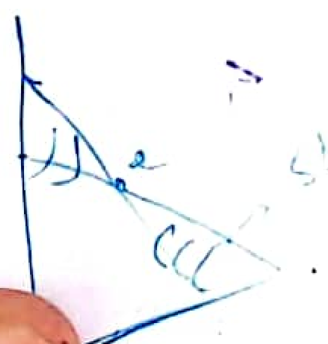
i. Pareto Improvement
ii. Nash Equilibrium
iii. Pay off Matrix
iv. Pareto Efficiency
v. Moral Hazard
vi. Free Rider problem

Answer the following questions. (3x10=30)

Q.2. Why does price leadership sometimes evolve in oligopolistic markets? Explain how the price leader determines a profit-maximizing price.

Q.3. Discuss the Inefficiency of Competition with Externalities.

Q.4. Explain the Stackelberg's model of duopoly.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – 2021

Roll No. ....

Paper: Advanced Mathematical Economics **(Re-Conduct)** **Time: 2 Hrs. 45 Min.** **Marks: 50**
Course Code: ECON-307-A **Part – II**

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions: **(5x4=20)**

I. Kuhn Tucker Conditions ✓

II. Concept of Primal and Dual ✓ ✓

III. Two Variable Phase Diagrams

IV. Calculus of Variations

V. The Maximum Principal

Answer the following questions. **(3x10=30)**

Q. 3. Find the Particular Solution of: **(5+5)**

i. $y'''(t) + 3y''(t) = \frac{1}{3}$

ii. $y'''(t) + y''(t) + 3y'(t) = 1$

✓**Q.4.** Apply Routh Theorem and find the dynamic stability of the time-path (without calculating the characteristic roots): $y'''(t) - 2y''(t) - y'(t) + 2y = 4$ **(10)**

Q. 5. Test the convergence of the path of $y_{t+2} + 3y_{t+1} + 2y_t = 12$, using Schur Theorem.

Book. **(10)**

$y'''(t) + 3y''(t) = \frac{1}{3}$

y_p of 3^{rd} order d equation is as: $y_p = 1$

$y_p = \frac{\frac{1}{3}}{\frac{1}{3}}$

$y_p = 1$

The characteristic equation of the above equation is: $r^3 + 3r^2 \cancel{r} = 0$

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – 2021

Roll No. ....

Paper: Advanced Mathematical Economics (Re-Conduct)
Course Code: ECON-307-A
Part – II
Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions: (5x4=20)

I. Kuhn Tucker Conditions

II. Concept of Primal and Dual

III. Two Variable Phase Diagrams

IV. Calculus of Variations

V. The Maximum Principal

Answer the following questions. (3x10=30)

Q. 3. Find the Particular Solution of: (5+5)

i. $y'''(t) + 3y''(t) = \frac{1}{3}$

ii. $y'''(t) + y''(t) + 3y'(t) = 1$

Q.4. *Apply Routh Theorem* and find the dynamic stability of the time-path (without calculating the *characteristic roots*): $y'''(t) - 2y''(t) - y'(t) + 2y = 4$ (10)

Q. 5. *Test the convergence of the path of* $y_{t+2} + 3y_{t+1} + 2y_t = 12$, using Schur Theorem. (10)

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – 2021

Roll No. ....

Paper: Advanced Mathematical Economics

Time: 2 Hrs. 45 Min. Marks: ...

Course Code: ECON-307 A Part – II

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following: (5x4=20)

I. Find the roots of the equation $3x^2 - x + 1 = 0$

II. What are exact differential equations?

III. Briefly explain Routh Theorem.

IV. Explain complex numbers

V. Solve $\frac{dy}{dt} + 4y = 2$ where $y(0) = 4$.

Give brief answers of the followings. (3x10=30)

Q. 3. Find the solution of $y''(t) + 9y'(t) - 10y = 5, y(0) = 6, y'(0) = 8\frac{1}{2}$

(10)

Q. 4. Explain the concept of linear programming with reference to its application in economics.

(10)

Q. 5. Solve the following difference equation and verify your answer: (10)

$y_{t+2} + 6y_{t+1} + 9y_t = 4$ where $y_0 = 1$ and $y_1 = 3$.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Spring 2022

Roll No. ....

Paper: Advanced Mathematical Economics **Course Code: ECON-307-A** **Time: 3 Hrs. Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. **(6x5=30)**

I. Find Particular Solution of: $y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$

II. Find characteristic roots of: $(r + 2)(r + 2)(r^2 + 2r + 2) = 0$

III. Write complementary function of characteristic equation $(r + 2)(r + 2)(r^2 + 2r + 2) = 0$.

IV. Explain Routh Theorem.

V. Explain the concept of Primal and Dual.

VI. Explain use of Linear Programming in Economics.

Answer the following questions. **(3x10=30)**

Q. 2. Test convergence of the path of $y_{t+2} + 2y_{t+1} + 2y_t = 12$, using Schur Theorem.

Q. 3. Explain Dynamic Input-output Models with example(s).

Q. 4. Solve the model: $X_{t+1} + \frac{1}{2}y_{t+1} - \frac{1}{7}y_t = 2$, using Routh Theorem.